MACHINE LEARNING PROJECT · P11

FC 24 Position Finder

Forward, midfielder or defender? Classifying EA Sports FC 24 players with a Random Forest

Name	KANARA JATIN JIJI
Roll No.	F10
Enrollment No.	24004501210126
Division	E
Project ID + title	P11 · FC 24 Position Finder (Sports)
Model / main metric	Random Forest / Macro F1
Final result (test set)	Macro F1 0.815 · Accuracy 0.818 · AUC 0.947

1. Problem statement

Using EA Sports FC 24 player attributes (pace, shooting, passing, dribbling, defending, physic and the detailed skills), classify every **outfield** player into one of three position groups: **FWD**, **MID** or **DEF**. This is a **multiclass classification** problem. A coach or scout could use it to suggest the best position for a player from his ratings alone.

2. Dataset description

- **Source:** Kaggle, “EA Sports FC 24 complete player dataset” (stefanoleone992), file `male_players.csv`, filtered to `fifa_version == 24`: **18,350 players × 109 columns**.
- **Target (Y):** first entry of `player_positions` → FWD = ST, CF, LW, RW, LF, RF · MID = CM, CDM, CAM, LM, RM · DEF = CB, LB, RB, LWB, RWB. 2,045 goalkeepers removed.
- **Classes:** MID 6,673 (40.9%) · DEF 6,174 (37.9%) · FWD 3,458 (21.2%) = 16,305 players. **Not balanced** (FWD is about half the size of MID), so Macro F1 is the main metric and the split uses `stratify=y`.

Data dictionary (grouped)

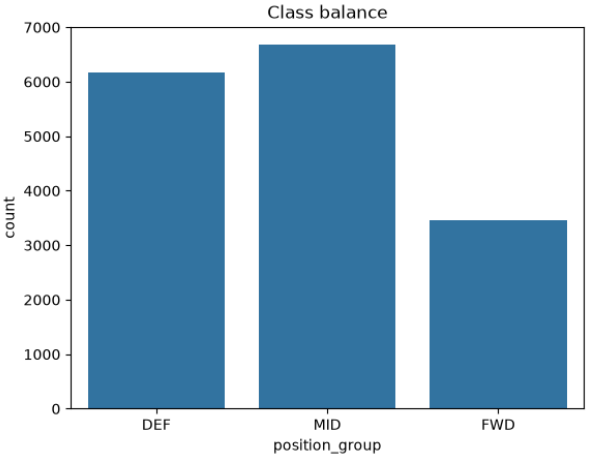
Column(s)	Meaning	Type	Keep / drop
`short_name / long_name`	Player name	Text	Drop (name, used only for search)
`player_positions`	Positions, first = main	Text	Used to make the target, then dropped (leak)
`pace, shooting, passing, dribbling, defending, physic`	6 main face ratings (0–99)	Number	Keep
`attacking_* (crossing, finishing, heading, short passing, volleys)`	Attacking sub-ratings	Number	Keep
`skill_* (dribbling, curve, fk_accuracy, long_passing, ball_control)`	Skill sub-ratings	Number	Keep
`movement_* / power_*`	Speed, agility, balance, shot power, jumping, stamina, strength, long shots	Number	Keep
`mentality_* / defending_*`	Aggression, interceptions, positioning, vision, penalties, composure, marking, tackles	Number	Keep
`goalkeeping_*`	GK ratings	Number	Keep (removed later by SelectKBest)
`age, height_cm, weight_kg, weak_foot, skill_moves, international_reputation`	Profile	Number	Keep
`preferred_foot, work_rate`	Foot, attack/defence work rate	Text	One-hot encoded
`overall, potential, ls…gk (27), club/nation_position`	Overall and position ratings	Number/Text	Drop: leakage
`player_id, player_url, *_id, club_name, league_name, dob, value_eur, wage_eur …`	IDs, links, names, money; goalkeeping_speed (empty)	Mixed	Drop: no player-skill meaning / empty

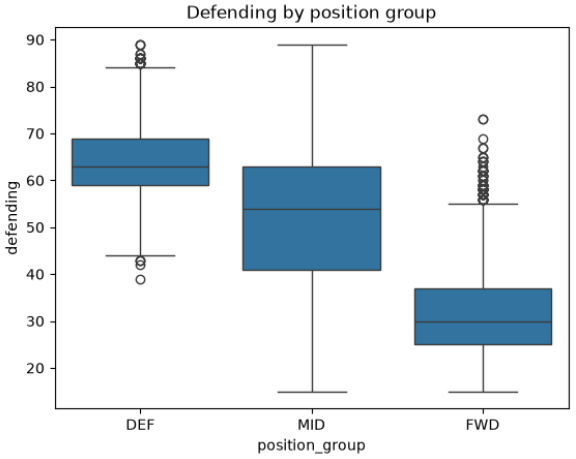
The notebook has the full 109-column dictionary (Step 3).

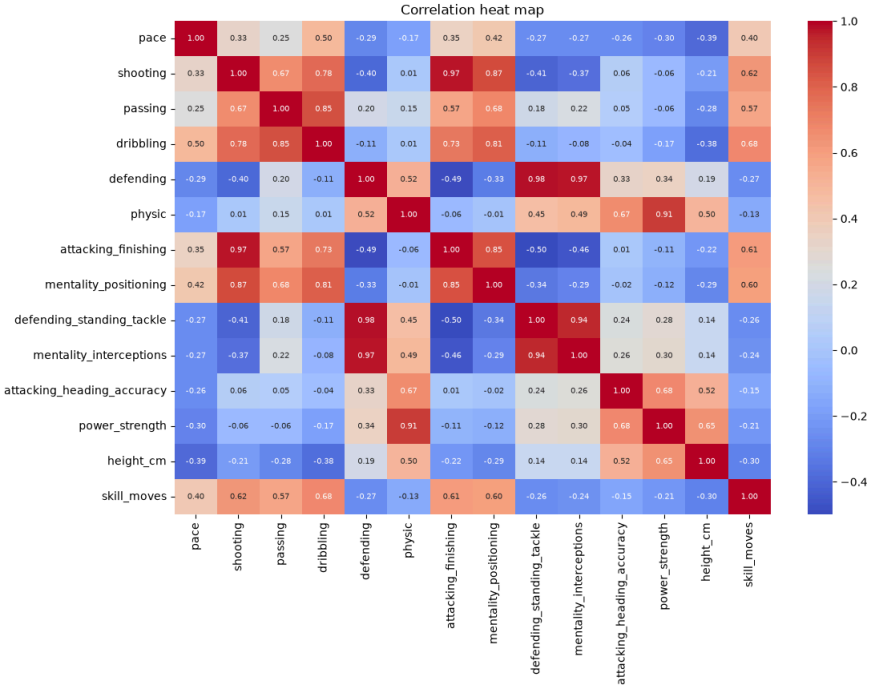
3. Cleaning

- Duplicates: 0 found. Missing values: median for number columns, mode for text columns.
- Dropped: IDs and links, names and text, money/club columns, columns that are the same for every row, `goalkeeping_speed` (empty), and **leaking columns** (player_positions, club/nation position, overall, potential and the 27 position ratings) that would give a fake ~100%.
- New feature: `attack_defence_gap = (shooting + dribbling)/2 − defending` (FWD +33.2, MID +10.2, DEF −14.8). One-hot encoding → 56 columns → **SelectKBest(f_classif, k = 15)**.

4. EDA: best charts and insights







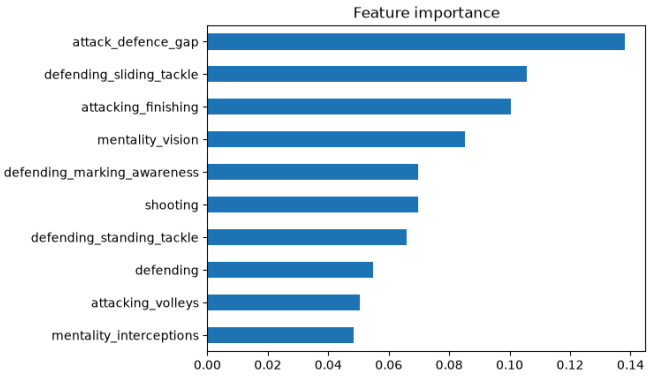
1. **Defending separates the groups best:** 63.7 (DEF), 51.8 (MID), 32.0 (FWD).
2. **Forwards shoot best:** shooting 64.5 (FWD), 57.5 (MID), 40.6 (DEF).
3. **Midfielders pass best** (62.4) and sit in the middle on attack and defence.
4. **Attack and defence go opposite ways:** defending vs finishing −0.49; defending, standing tackle and interceptions are 0.97–0.98 alike.
5. **Defenders are tallest** (183.0 cm) and most physical (68.5); forwards are fastest (pace 72.0).

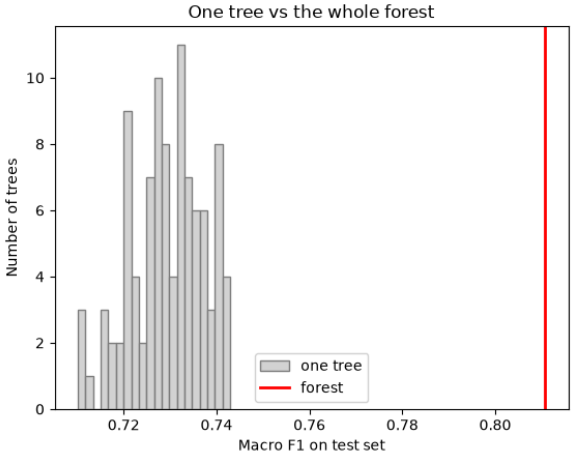
5. Model and settings tried

Random Forest (scikit-learn), 80/20 split with `stratify=y`, `random_state=42`, 15 features chosen by SelectKBest:

`attack_defence_gap, defending_sliding_tackle, defending_standing_tackle, attacking_finishing, defending, defending_marking_awareness, mentality_interceptions, shooting, attacking_volleys, power_long_shots, mentality_penalties, mentality_positioning, mentality_vision, power_shot_power, skill_dribbling`

Baseline (default, 100 trees), then **n_estimators = 50 / 100 / 200** and **max_depth = None / 5 / 10**. Each setting is scored on the test set with accuracy, macro precision, recall and F1, specificity and AUC (one-vs-rest).





Top 5 features (100-tree forest): attack_defence_gap 0.138, defending_sliding_tackle 0.106, attacking_finishing 0.100, mentality_vision 0.085, defending_marking_awareness 0.070. **Forest vs one tree:** single trees scored macro F1 0.710–0.743 (avg 0.729); the forest's vote scored 0.811.

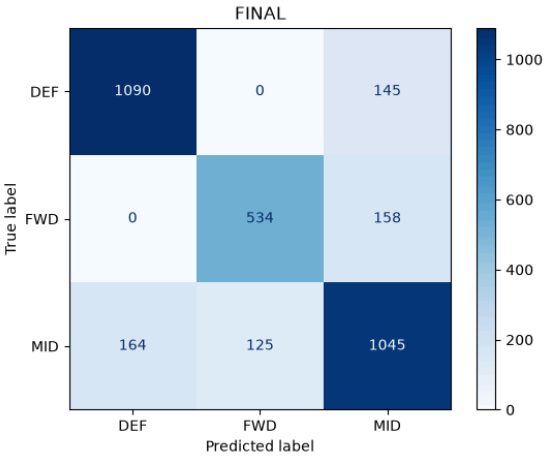
6. Results

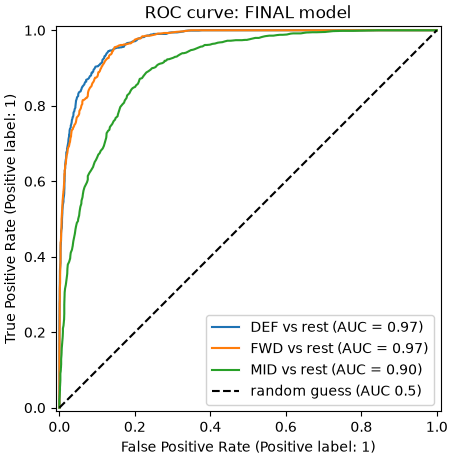
Setting	Accuracy	Precision	Recall	Macro F1	Specificity	AUC
Baseline	0.813	0.812	0.809	0.811	0.902	0.945
n_estimators=50	0.810	0.808	0.808	0.808	0.901	0.943
n_estimators=100	0.813	0.812	0.809	0.811	0.902	0.945
n_estimators=200 ★	**0.818**	**0.818**	**0.813**	**0.815**	**0.904**	**0.947**
max_depth=None	0.813	0.812	0.809	0.811	0.902	0.945
max_depth=5	0.782	0.778	0.784	0.781	0.886	0.928
max_depth=10	0.812	0.813	0.807	0.810	0.901	0.943

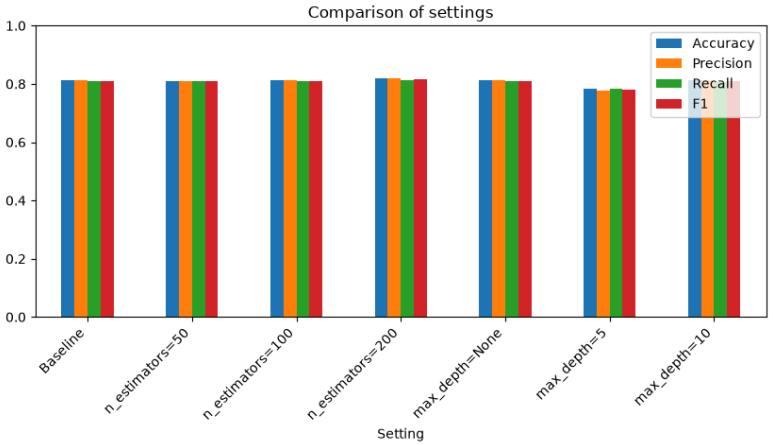
★ **Final model: n_estimators = 200, max_depth = None**: highest macro F1 and best on every other metric.

6. Results (continued): final model

0.815	0.818	0.818	0.813	0.904	0.947
Macro F1	Accuracy	Precision	Recall	Specificity	AUC







Confusion matrix: 2,669 of 3,261 test players correct. DEF recall 0.883; FWD precision 0.810, recall 0.772, specificity 0.951; MID precision 0.775. No defender was predicted as a forward and no forward as a defender: all mistakes involve MID.

3 new inputs: winger → FWD (72.5%), playmaker → MID (86.5%), centre-back → DEF (99.0%). All three make sense.

7. Conclusion and what I learned

- From FC 24 ratings alone the Random Forest tells forwards, midfielders and defenders apart with **macro F1 0.815** and AUC 0.947 on unseen players.
- Leaking columns must go first; one well-made feature (attack_defence_gap) became the strongest; macro F1 is the right choice for imbalanced classes; a forest is much more stable than one tree.
- The model also runs as a web app (GitHub Pages) and a Streamlit app using the same 15 features and the same final forest.